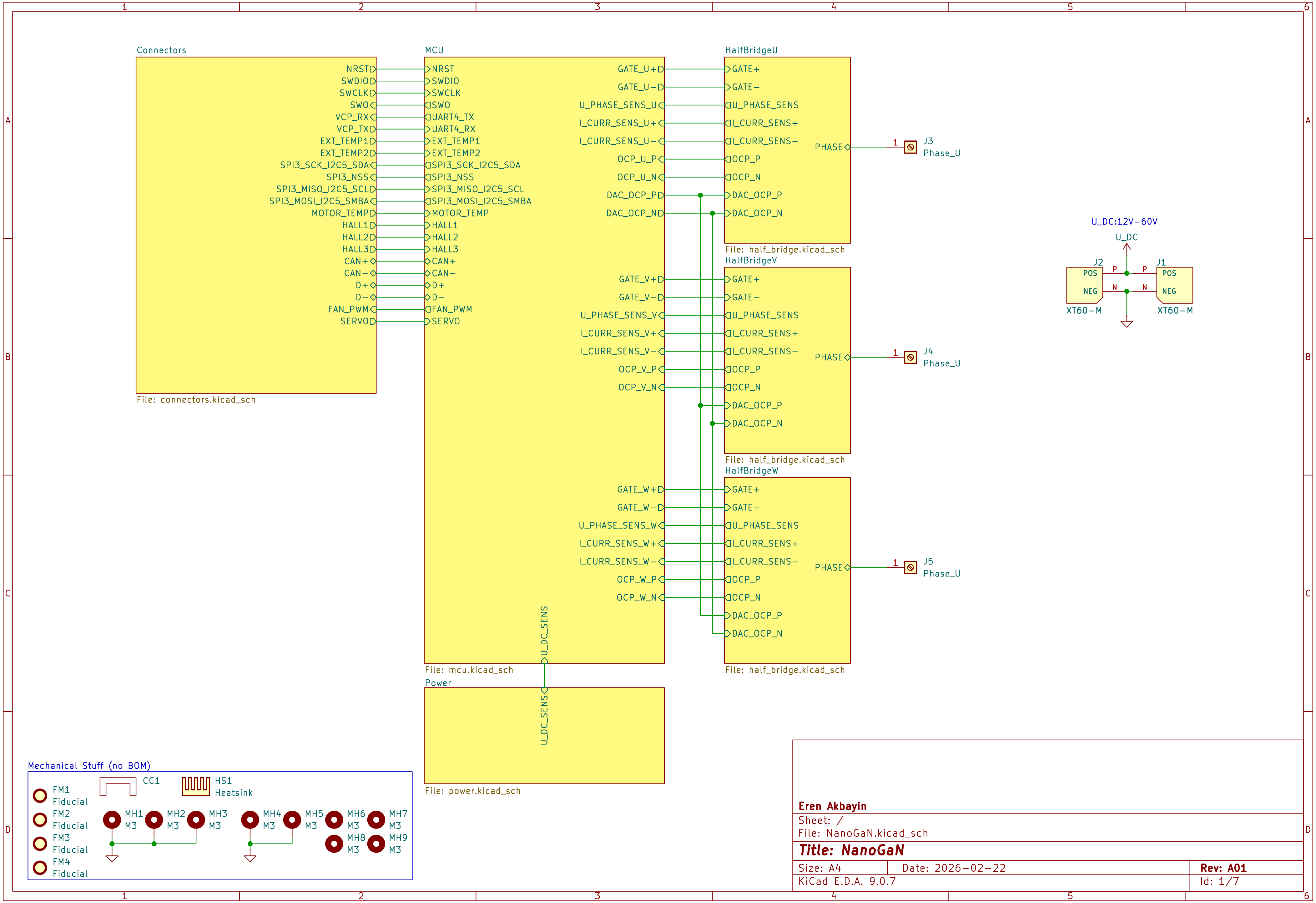
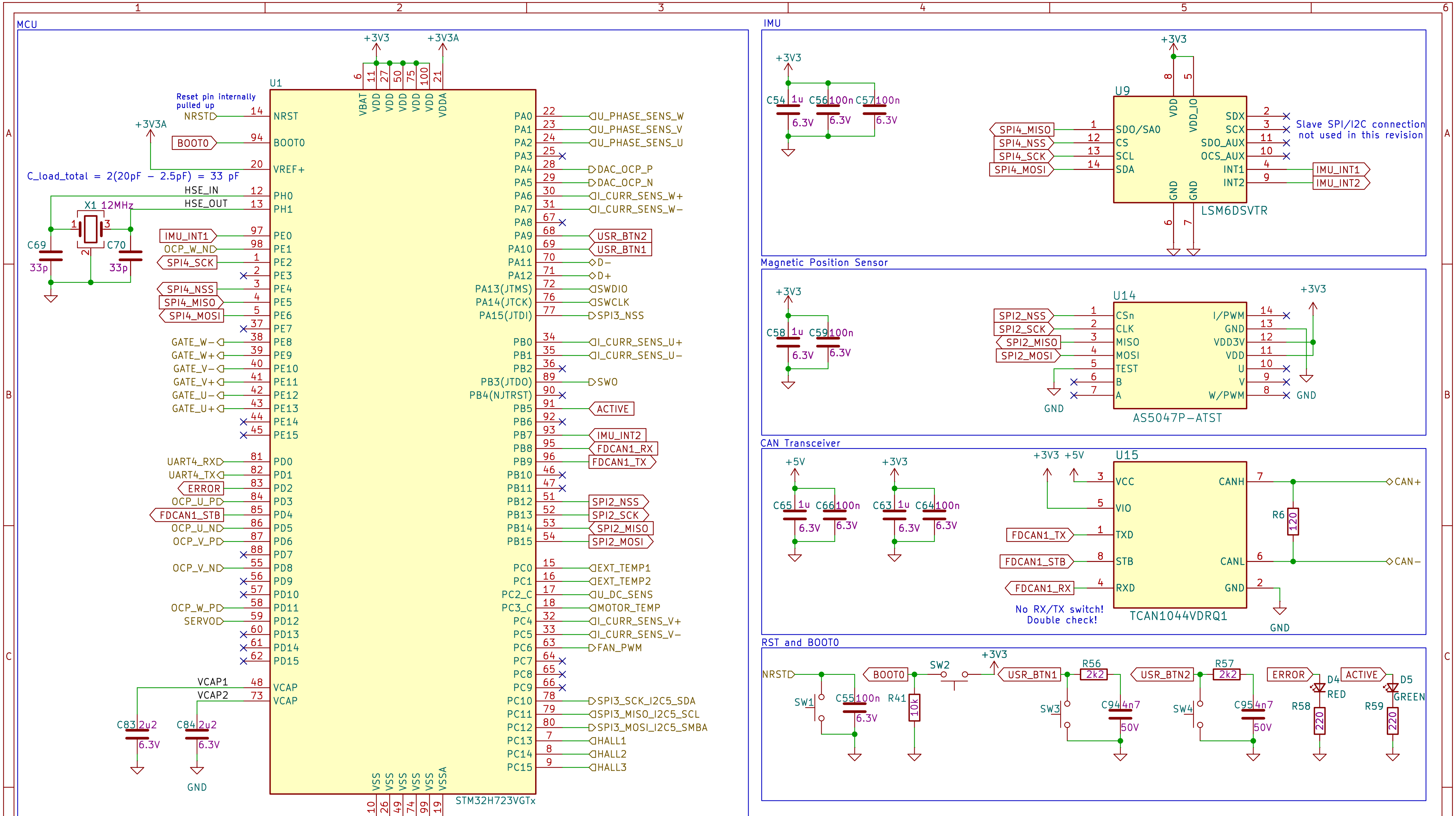

Connectors
NRST
SWDIO
SWCLK
SWO
VCP_RX
VCP_TX
EXT_TEMP1
EXT_TEMP2
SPI3_SCK_I2C5_SDA
SPI3_NSS
SPI3_MISO_I2C5_SCL
SPI3_MOSI_I2C5_SMBA
MOTOR_TEMP
HALL1
HALL2
HALL3
CAN+
CAN−
D+
D−
FAN_PWM
SERVO
File: connectors.kicad_sch
MCU
NRST
SWDIO
SWCLK
SWO
UART4_TX
UART4_RX
EXT_TEMP1
EXT_TEMP2
SPI3_SCK_I2C5_SDA
SPI3_NSS
SPI3_MISO_I2C5_SCL
SPI3_MOSI_I2C5_SMBA
MOTOR_TEMP
HALL1
HALL2
HALL3
CAN+
CAN−
D+
D−
FAN_PWM
SERVO
GATE_U+
GATE_U−
U_PHASE_SENS_U
I_CURR_SENS_U+
I_CURR_SENS_U−
OCP_U_P
OCP_U_N
DAC_OCP_P
DAC_OCP_N
GATE_V+
GATE_V−
U_PHASE_SENS_V
I_CURR_SENS_V+
I_CURR_SENS_V−
OCP_V_P
OCP_V_N
GATE_W+
GATE_W−
U_PHASE_SENS_W
I_CURR_SENS_W+
I_CURR_SENS_W−
OCP_W_P
OCP_W_N
U_DC_SENS
File: mcu.kicad_sch
HalfBridgeU
GATE+
GATE−
U_PHASE_SENS
I_CURR_SENS+
I_CURR_SENS−
OCP_P
OCP_N
DAC_OCP_P
DAC_OCP_N
PHASE
File: half_bridge.kicad_sch
HalfBridgeV
GATE+
GATE−
U_PHASE_SENS
I_CURR_SENS+
I_CURR_SENS−
OCP_P
OCP_N
DAC_OCP_P
DAC_OCP_N
PHASE
File: half_bridge.kicad_sch
HalfBridgeW
GATE+
GATE−
U_PHASE_SENS
I_CURR_SENS+
I_CURR_SENS−
OCP_P
OCP_N
DAC_OCP_P
DAC_OCP_N
PHASE
File: half_bridge.kicad_sch
1 J3 Phase_U
1 J4 Phase_U
1 J5 Phase_U
U_DC:12V−60V
U_DC
J2 POS NEG XT60−M
J1 POS NEG XT60−M
P P N N
Power
U_DC_SENS
File: power.kicad_sch
Mechanical Stuff (no BOM)
FM1 Fiducial
FM2 Fiducial
FM3 Fiducial
FM4 Fiducial
CC1
HS1 Heatsink
MH1 M3
MH2 M3
MH3 M3
MH4 M3
MH5 M3
MH6 M3
MH7 M3
MH8 M3
MH9 M3
Eren Akbayin
Sheet: /
File: NanoGaN.kicad_sch
Title: NanoGaN
Size: A4
Date: 2026−02−22
Rev: A01
KiCad E.D.A. 9.0.7
Id: 1/7



Eren Akbayin

Sheet: /MCU/

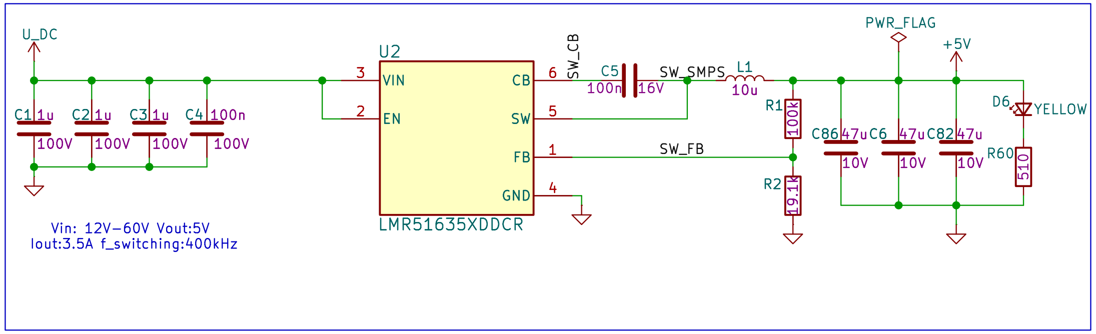
File: mcu.kicad_sch

Title: NanoGaN

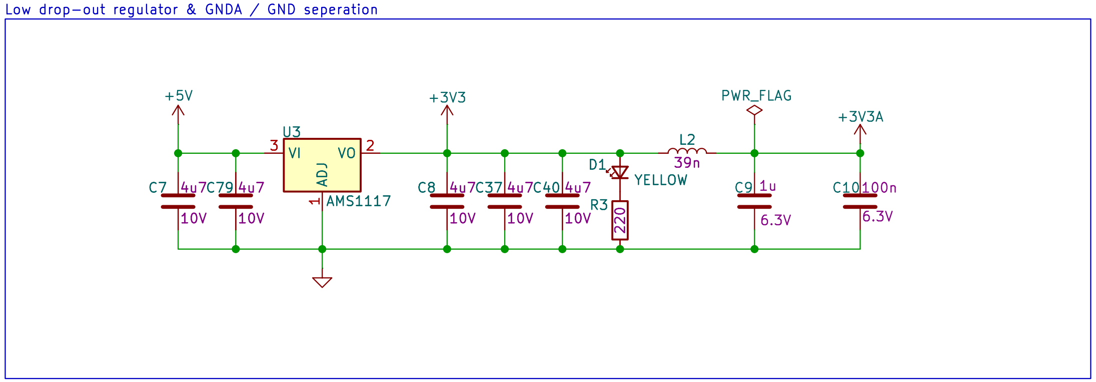
Size: A4 | Date: 2026-02-22 | Rev: A01

KiCad E.D.A. 9.0.7 | Id: 2/7

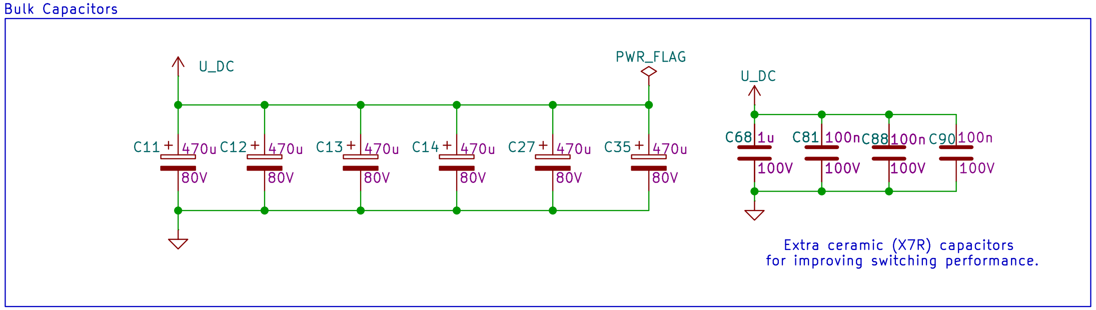
Buck Converter



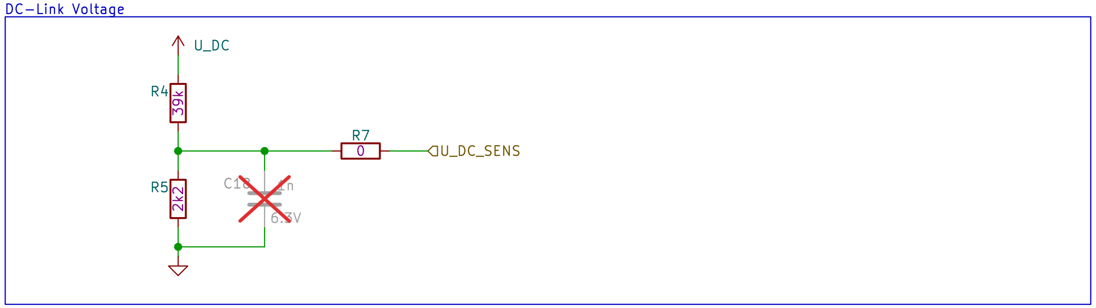
Low drop-out regulator & GNDA / GND seperation

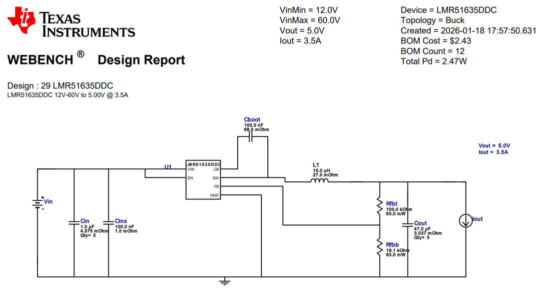


Bulk Capacitors



DC-Link Voltage





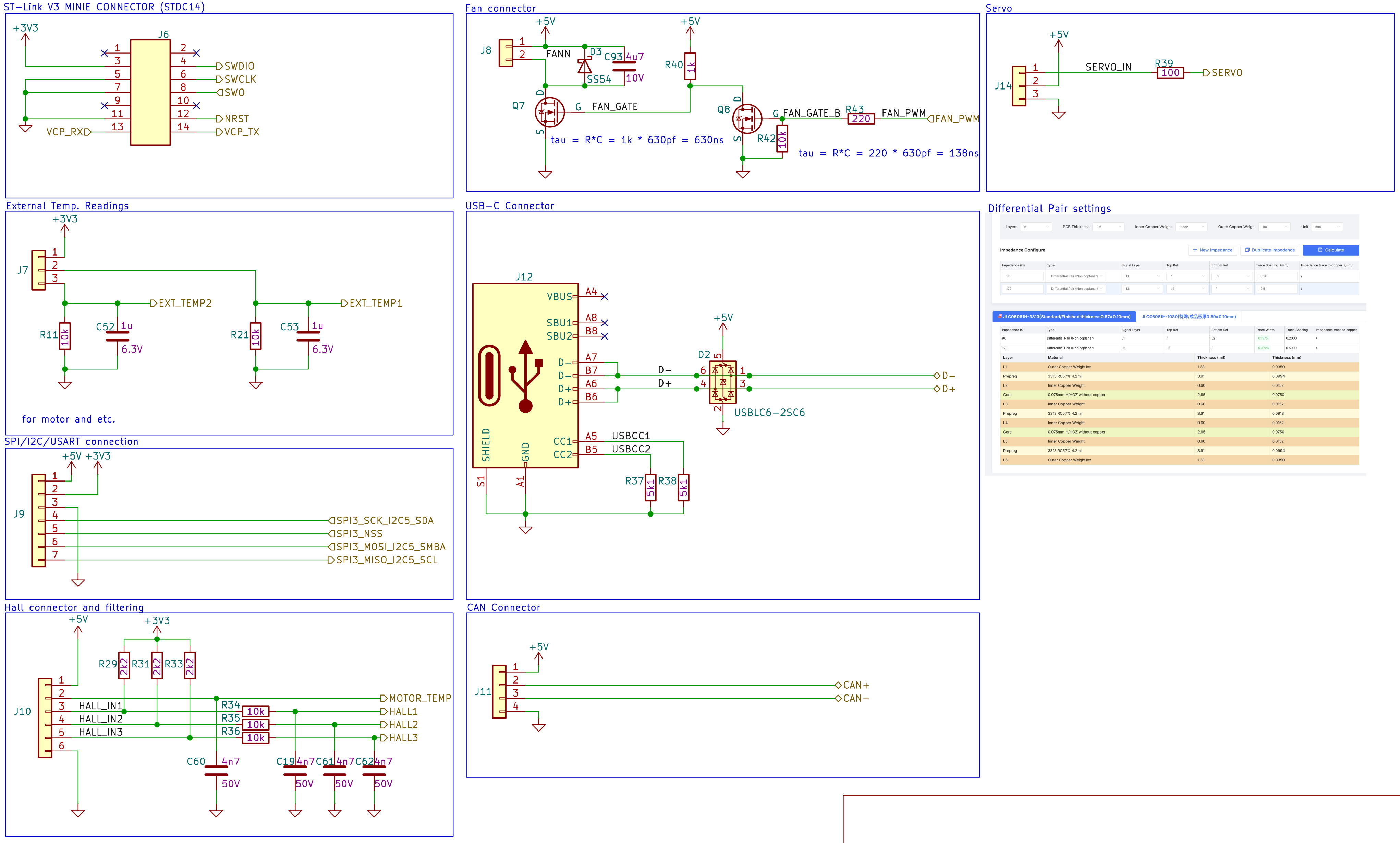
Eren Akbayin

Sheet: /Power/

File: power.kicad_sch

Title: NanoGaN

Size: A4	Date: 2026-02-22	**Rev: A01**
KiCad E.D.A. 9.0.7		Id: 3/7

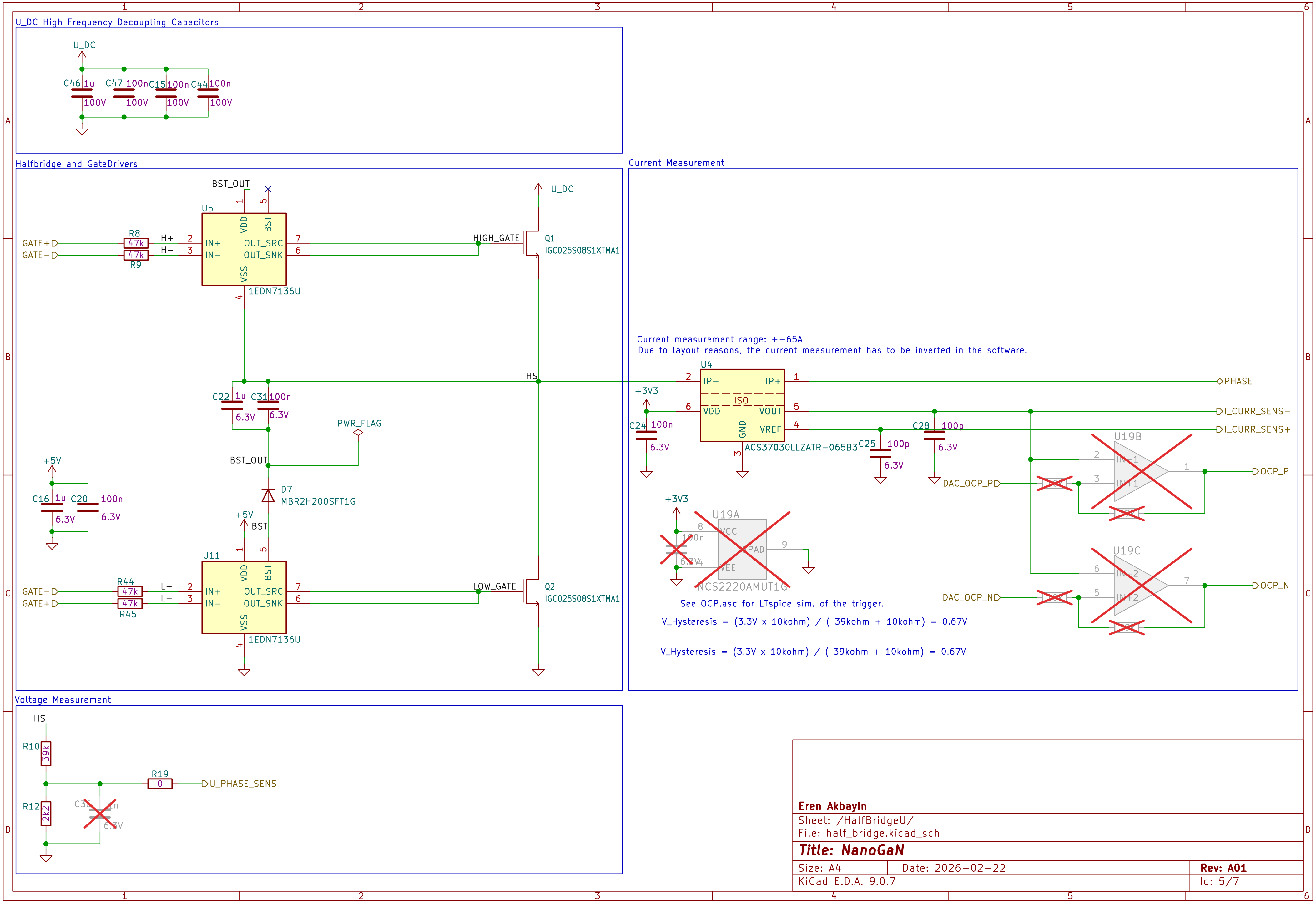


Eren Akbayin

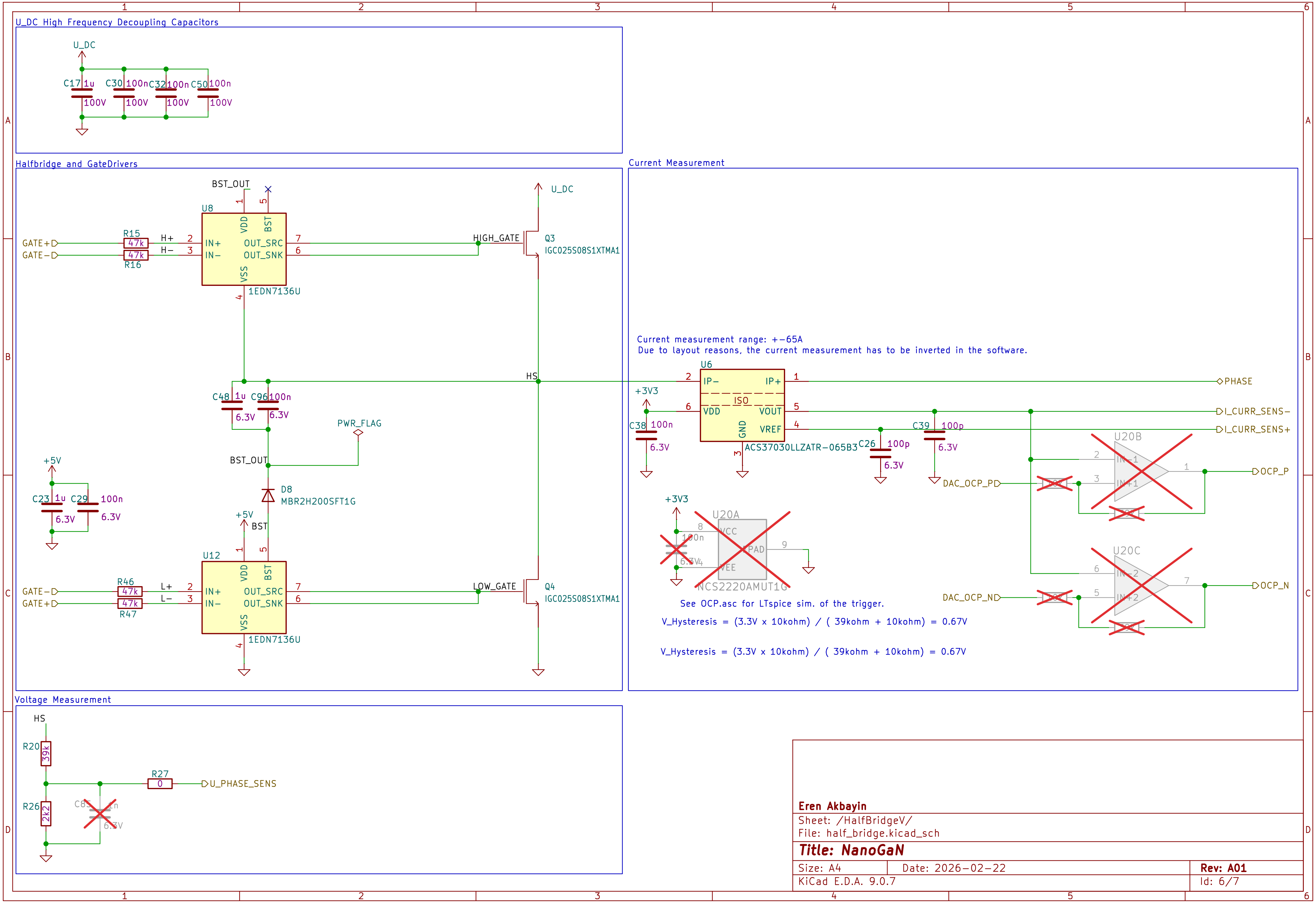
Sheet: /Connectors/
File: connectors.kicad_sch

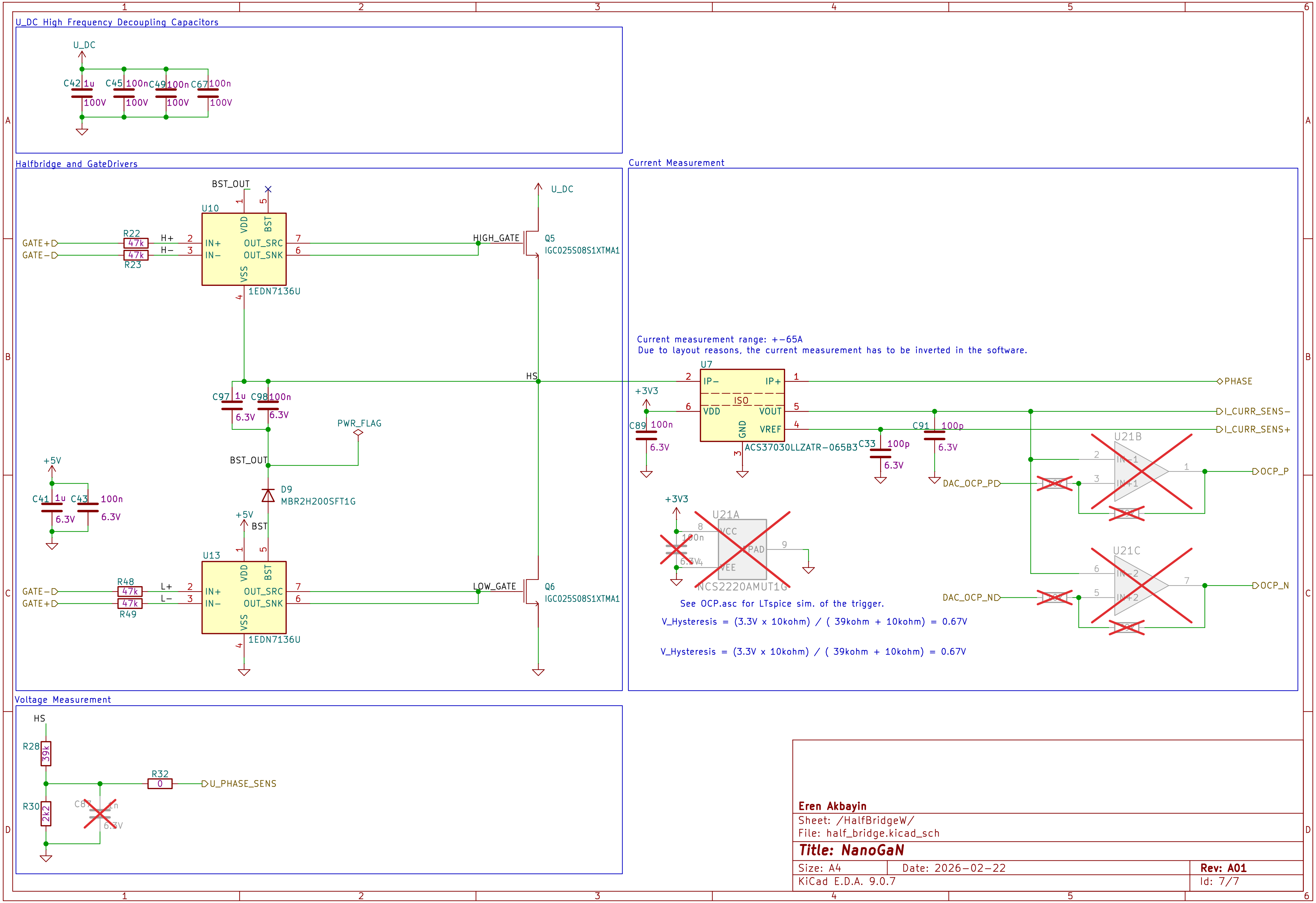
Title: NanoGaN

Size: A4	Date: 2026-02-22	**Rev: A01**
KiCad E.D.A. 9.0.7		Id: 4/7



U_DC High Frequency Decoupling Capacitors
U_DC
C46 1u 100V
C47 100n 100V
C15 100n 100V
C44 100n 100V
Halfbridge and GateDrivers
BST_OUT
U5
1EDN7136U
VDD
BST
IN+
IN−
OUT_SRC
OUT_SNK
VSS
GATE+
GATE−
R8 47k
R9 47k
H+
H−
HIGH_GATE
Q1 IGC025S08S1XTMA1
U_DC
HS
C22 1u 6.3V
C31 100n 6.3V
PWR_FLAG
BST_OUT
D7 MBR2H200SFT1G
+5V
BST
C16 1u 6.3V
C20 100n 6.3V
U11
1EDN7136U
R44 47k
R45 47k
L+
L−
LOW_GATE
Q2 IGC025S08S1XTMA1
Current Measurement
Current measurement range: +−65A
Due to layout reasons, the current measurement has to be inverted in the software.
U4
ACS37030LLZATR-065B3
IP−
IP+
ISO
VDD
VOUT
GND
VREF
PHASE
+3V3
C24 100n 6.3V
C25 100p 6.3V
C28 100p 6.3V
I_CURR_SENS−
I_CURR_SENS+
U19B
DAC_OCP_P
OCP_P
U19A
NCS2220AMUT1G
VCC
VEE
PAD
U19C
DAC_OCP_N
OCP_N
See OCP.asc for LTspice sim. of the trigger.
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
Voltage Measurement
HS
R10 39k
R12 2k2
R19 0
U_PHASE_SENS
Eren Akbayin
Sheet: /HalfBridgeU/
File: half_bridge.kicad_sch
Title: NanoGaN
Size: A4
Date: 2026-02-22
Rev: A01
KiCad E.D.A. 9.0.7
Id: 5/7


U_DC High Frequency Decoupling Capacitors
U_DC
C17 1u 100V
C30 100n 100V
C32 100n 100V
C50 100n 100V
Halfbridge and GateDrivers
BST_OUT
U8
VDD
BST
GATE+
GATE−
R15 47k
R16 47k
H+
H−
IN+
IN−
OUT_SRC
OUT_SNK
VSS
1EDN7136U
HIGH_GATE
Q3
IGC025S08S1XTMA1
U_DC
HS
C48 1u 6.3V
C96 100n 6.3V
BST_OUT
PWR_FLAG
D8
MBR2H200SFT1G
+5V
C23 1u 6.3V
C29 100n 6.3V
+5V
BST
U12
GATE−
GATE+
R46 47k
R47 47k
L+
L−
IN+
IN−
OUT_SRC
OUT_SNK
VSS
1EDN7136U
LOW_GATE
Q4
IGC025S08S1XTMA1
Current Measurement
Current measurement range: +-65A
Due to layout reasons, the current measurement has to be inverted in the software.
U6
IP−
IP+
ISO
VDD
VOUT
VREF
GND
ACS37030LLZATR-065B3
PHASE
+3V3
C38 100n 6.3V
C26 100p 6.3V
C39 100p 6.3V
I_CURR_SENS−
I_CURR_SENS+
U20B
DAC_OCP_P
OCP_P
+3V3
U20A
VCC
VEE
PAD
NCS2220AMUT1G
U20C
DAC_OCP_N
OCP_N
See OCP.asc for LTspice sim. of the trigger.
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
Voltage Measurement
HS
R20 39k
R27 0
U_PHASE_SENS
R26 2k2
Eren Akbayin
Sheet: /HalfBridgeV/
File: half_bridge.kicad_sch
Title: NanoGaN
Size: A4
Date: 2026-02-22
Rev: A01
KiCad E.D.A. 9.0.7
Id: 6/7


U_DC High Frequency Decoupling Capacitors
U_DC
C42 1u 100V
C45 100n 100V
C49 100n 100V
C67 100n 100V
Halfbridge and GateDrivers
BST_OUT
U10
VDD
BST
GATE+
GATE−
R22 47k
R23 47k
H+
H−
IN+
IN−
OUT_SRC
OUT_SNK
VSS
1EDN7136U
HIGH_GATE
Q5
IGC025S08S1XTMA1
U_DC
HS
C97 1u 6.3V
C98 100n 6.3V
PWR_FLAG
BST_OUT
D9
MBR2H200SFT1G
+5V
BST
+5V
C41 1u 6.3V
C43 100n 6.3V
U13
GATE−
GATE+
R48 47k
R49 47k
L+
L−
LOW_GATE
Q6
IGC025S08S1XTMA1
1EDN7136U
Current Measurement
Current measurement range: +−65A
Due to layout reasons, the current measurement has to be inverted in the software.
U7
IP−
IP+
ISO
VDD
VOUT
VREF
GND
ACS37030LLZATR−065B3
PHASE
+3V3
C89 100n 6.3V
C33 100p 6.3V
C91 100p 6.3V
I_CURR_SENS−
I_CURR_SENS+
U21B
DAC_OCP_P
OCP_P
+3V3
U21A
VCC
PAD
VEE
NCS2220AMUT1G
U21C
DAC_OCP_N
OCP_N
See OCP.asc for LTspice sim. of the trigger.
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
V_Hysteresis = (3.3V x 10kohm) / (39kohm + 10kohm) = 0.67V
Voltage Measurement
HS
R28 39k
R30 2k2
R32 0
C87
U_PHASE_SENS
Eren Akbayin
Sheet: /HalfBridgeW/
File: half_bridge.kicad_sch
Title: NanoGaN
Size: A4
Date: 2026−02−22
KiCad E.D.A. 9.0.7
Rev: A01
Id: 7/7